

**Architecting a Deterministic Governed
Process CPU:
The Chatman Engine S1-S6 Pipeline and
PDDL-to-POWL Projection**

Sean Chatman

July 10, 2026

Abstract

This thesis presents the architecture and verification of the Chatman Engine v26.7.9, a fully governed process CPU designed for deterministic agentic execution. We introduce the S1-S6 pipeline, leveraging RDF/Oxigraph semantic authority, rigorous 9-hash receipt generation, and a complete PDDL-to-POWL trace admission. By replacing positional event IDs with stable ordinals in OCEL evidence and transitioning to strictly typed RDF acceptance fixtures, we achieve byte-identical receipt digests across executions. Furthermore, an integrated 256-entry branchless admission table under a bounded 8-logical-tick cycle minimizes hot-path variance. We outline the architectural innovations, empirical determinism benchmarks, and mathematical models—spanning algebraic mappings, differential calculus optimizations, and geometric manifolds—that enable secure and immutable execution logs.

Contents

1	Introduction	2
2	The S1-S6 Pipeline: Algebraic Formalization	3
3	Deterministic OCEL Evidence: Differential Optimization	4
4	Governance and Typestate Integrity: Geometric Boundaries	5
5	Conclusion	6

Chapter 1

Introduction

The necessity for predictable, deterministic execution in agentic systems drives the development of the Chatman Engine. Contemporary AI pipelines often suffer from non-deterministic variance in event logging and artifact generation. We present an architecture—the S1-S6 pipeline—that strictly governs process transitions. By pinning `oxigraph/oxrdf` dependencies and executing within an isolated `RDFTriple8` hot path, we establish absolute control over process state boundaries.

Chapter 2

The S1-S6 Pipeline: Algebraic Formalization

The S1-S6 pipeline operates over six distinct stages. We can formalize this pipeline algebraically as a composite transformation from the state space $\mathcal{S}$ to the receipt space $\mathcal{R}$. Let Ψ denote the full process evaluation:

$$\Psi : \mathcal{S} \rightarrow \mathcal{R} \tag{2.1}$$

The transformation Ψ is composed of six isolated functions, $\Psi(x) = (f_6 \circ f_5 \circ f_4 \circ f_3 \circ f_2 \circ f_1)(x)$, defined as:

- f_1 : Snapshot derivation mapping via RDFC-1.0 hashing.
- f_2 : OWL closure computation.
- f_3 : PDDL plan synthesis projection.
- f_4 : POWL v2 trace admission.
- f_5 : Knowledge hooks resolution.
- f_6 : 9-hash `ProcessReceiptEnvelope` generation.

The `RDFTriple8` hot path utilizes a 256-entry branchless admission table, modeled as a surjective homomorphism:

$$\mathcal{A} : \mathbb{F}_2^8 \rightarrow \mathbb{Z}_{256} \tag{2.2}$$

where $\mathbb{F}_2^8$ represents the 8-bit Chatman Constant constraint vector, establishing an absolute $O(1)$ lookup bound for trace admissions.

Chapter 3

Deterministic OCEL Evidence: Differential Optimization

A critical innovation in v26.7.9 is the transition from positional event IDs to stable ordinals in Object-Centric Event Logs (OCEL). Parallel test execution previously induced race conditions that corrupted digest determinism.

We define the variance of the system $V(t)$ over execution time t . By anchoring ordinals, we strictly enforce that the differential rate of variance is absolute zero:

$$\lim_{\Delta t \rightarrow 0} \frac{V(t + \Delta t) - V(t)}{\Delta t} = \frac{dV}{dt} = 0 \quad (3.1)$$

To achieve the optimal bounded path execution, we minimize the latency penalty $\mathcal{L}(\theta, t)$ integrated over the critical execution path spanning from t_0 to t_1 :

$$\min_{\theta \in \Theta} \int_{t_0}^{t_1} (\nabla \mathcal{L}(\theta, t) \cdot \mathbf{v}(t) + \lambda \|\theta\|^2) dt \quad (3.2)$$

This establishes a bounded integral demonstrating that execution cost is purely a function of the semantic payload, rendering it immune to multithreading context switching.

Chapter 4

Governance and Typestate Integrity: Geometric Boundaries

We document the implementation of a 29-variant typed **Refusal** enum, eliminating panic and unwrap vulnerabilities. Geometrically, we model the valid operational state space as a bounded convex manifold $\mathcal{M} \subset \mathbb{R}^n$.

Any valid process trace $\gamma(s)$ must lie strictly within the interior of $\mathcal{M}$. The process boundaries are enforced by a metric space distance function $d(x, y)$:

$$d(x, \partial\mathcal{M}) > \epsilon, \quad \forall x \in \gamma(s) \quad (4.1)$$

The 29-variant **Refusal** boundary $\partial\mathcal{M}$ acts as an absolute constraint surface. If an evaluation vector $\vec{v}$ intersects $\partial\mathcal{M}$, the transition is sealed and projected onto an orthogonal refusal tangent space:

$$T_x\mathcal{M} \oplus T_x^\perp\mathcal{M} = \mathbb{R}^n \quad (4.2)$$

This topological guarantee ensures that all execution paths are securely contained and verifiable via Gates A-E.

Chapter 5

Conclusion

The Chatman Engine establishes a new standard for determinism in governed process execution. The integration of rigorous algebraic mappings, optimal differential bounds, and topological constraints guarantee verifiable, trustless replays of automated tasks.